

From a logical analysis of the above definitions, it is clear that the definition of Statistics given by Croxton and Cowden is the most scientific and realistic one, and will form the subject-matter of our study.

1.3. IMPORTANCE AND SCOPE OF STATISTICS

In modern times, Statistics is viewed not as a mere device for collecting numerical data but as a means of developing sound techniques for their handling and analysis and drawing valid inferences from them. As such it is not confined to the affairs of the State but is intruding constantly into various diversified spheres of life - social as well as physical - such as biology, psychology, education, economics, business management, etc. It is hardly possible to enumerate even a single department of human activity where statistics does not creep in. It has rather become indispensable in all phases of human endeavour.

We now discuss briefly the importance of Statistics in some different sectors and disciplines.

1. Statistics and Planning. Statistics is indispensable to planning. In the modern age which is termed as *'the age of planning'*, almost all organisations in the government or managements of business are resorting to planning for efficient working and for formulating policy decisions. To achieve this end, the statistical data relating to production, consumption, prices, investment, income, expenditure, etc and various advance statistical techniques for handling and analysing such complex data are of paramount importance.

2. Statistics and Mathematics. Statistics is intimately related to and essentially dependent upon mathematics. The modern theory of statistics has its foundations on the theory of probability which in turn is a particular branch of more advanced mathematical theory of Measure and Intergration. Moreover, the developments of statistical techniques and theories for application to various sciences-social, physical and natural, are based on fitting different mathematical models to the observed data under certain assumptions and the whole process of such assumptions, analysis and testing is basically mathematical in character, making wide applications of mathematical tools of integration, differentiation, algebra, trigonometry, matrix theory and so on. The main stalwarts in the theory of modern Statistics namely Laplace, Bernoulli, Pascal, De-Moivre, Gauss, Chebychev, Kolmogorov, R.A. Fisher, S.N. Roy, R.C. Bose, H. Cramer to mention only a few of them were primarily talented and skilled mathematicians. Ever increasing role of mathematics in Statistics has resulted in the development of a new branch of Statistics called *'Mathematical Statistics.'* In the words of Connor:

'Statistics is a branch of applied mathematics which specialises in data.'

3. Statistics and Economics. Statistical data and techniques of statistical analysis have proved immensely useful in solving a variety of economic problems, such as wages, prices, consumption, production, distribution of income and wealth etc. Statistical tools like Index Numbers, Time Series Analysis, Demand Analysis and Forecasting Techniques are extensively used for efficient planning and economic development of a country. [For a detailed discussion on these topics, the reader may refer to the book *'Fundamentals of Applied Statistics'* by the same author.]

Hence, the average age of those retained in the industrial house is 47 years.

2.5.2. Merits and Demerits of Arithmetic Mean

Merits

1. It is rigidly defined.
2. It is easy to understand and easy to calculate.
3. It is based upon all the observations.
4. It is amenable to algebraic treatment. The mean of the composite series in terms of the means and sizes of the component series is given by :

$$\bar{x} = \frac{\sum_{i=1}^k n_i \bar{x}_i}{\sum_{i=1}^k n_i}$$

5. Of all the averages, arithmetic mean is affected least by fluctuations of sampling. This property is sometimes described by saying that arithmetic mean is a *stable* average.

Thus, we see that arithmetic mean satisfies all the properties laid down by Prof. Yule for an ideal average.

Demerits

1. It cannot be determined by inspection nor it can be located graphically.
2. Arithmetic mean cannot be used if we are dealing with qualitative characteristics which cannot be measured quantitatively; such as, intelligence, honesty, beauty, etc.
3. Arithmetic mean cannot be obtained if a single observation is missing or lost or is illegible unless we drop it out and compute the arithmetic mean of the remaining values.
4. Arithmetic mean is affected very much by extreme values. In case of extreme items, arithmetic mean gives a distorted picture of the distribution and no longer remains representative of the distribution.

5. Arithmetic mean may lead to wrong conclusions if the details of the data from which it is computed are not given. Let us consider the following marks obtained by two students A and B in three tests, viz., terminal test, half-yearly examination and annual examination respectively.

Marks	I Test	II Test	III Test	Average
A	50%	60%	70%	60%
B	70%	60%	50%	60%

Thus average marks obtained by each of the two students at the end of the year are 60%. If we are given the average marks alone we conclude that the level of intelligence of both the students at the end of the year is same. This is a fallacious conclusion since we find from the data that student A has improved consistently while student B has deteriorated consistently.

6. Arithmetic mean cannot be calculated if the extreme class is open, e.g., below 10 or above 90.

7. In extremely asymmetrical (skewed) distribution, usually arithmetic mean is not a suitable measure of location.

2.6.2. Merits and Demerits of Median

Merits

1. It is rigidly defined.
2. It is easily understood and is easy to calculate. In some cases it can be located merely by inspection.
3. It is not at all affected by extreme values.
4. It can be calculated for distributions with open-end classes.

Demerits

1. In case of even number of observations median cannot be determined exactly. We merely estimate it by taking the mean of two middle terms.
2. It is not based on all the observations. For example, the median of 10, 25, 50, 60 and 64 is 50. We can replace the observations 10 and 25 by any two values which are smaller than 50 and the observations 60 and 65 by any two values greater than 50, without affecting the value of median. This property is sometimes described by saying that median is *insensitive*.
3. It is not amenable to algebraic treatment.
4. As compared with mean, it is affected much by fluctuations of sampling.

Uses (i) Median is the only measure of central tendency which is not affected by extreme values.

2.7.2. Merits and Demerits of Mode

Merits	Demerits
1. Mode is readily comprehensible and easy to calculate. Like median, mode can be located in some cases merely by inspection. 2. Mode is not at all affected by extreme values. 3. Mode can be conveniently located even if the frequency distribution has class-intervals of unequal magnitude provided the modal class and the classes preceding and succeeding it are of the same magnitude. Open-end classes also do not pose any problem in the location of mode.	1. Mode is ill-defined. It is not always possible to find a clearly defined mode. In some cases, we may come across distributions with two modes. Such distributions are called <i>bi-modal</i>. If a distribution has more than two modes, it is said to be <i>multimodal</i>. 2. It is not based upon all the observations. 3. It is not capable of further mathematical treatment. 4. As compared with mean, mode is affected to a greater extent, by fluctuations of sampling.

How Mode is used to find the ideal size. e.g., in business